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Systems and related temperature calibration methods

Abstract

Systems and related temperature calibration methods. In accordance with a first implementation, an apparatus includes a flow cell interface, a temperature control device, an infrared sensor, and a controller. The flow cell interface includes a flow cell support and the temperature control device is for the flow cell support. The controller is to command the temperature control device to cause the flow cell support to achieve a temperature value, cause the infrared sensor to measure an actual temperature value of the flow cell support, and calibrate the temperature control device based on a difference between the commanded temperature value and the actual temperature value.

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Background/Summary

RELATED APPLICATION SECTION (1) This application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/293,294, filed Dec. 23, 2021, the content of which is incorporated by reference herein in its entirety and for all purposes.

BACKGROUND

(1) Instruments such as sequencing instruments may include temperature controlled components.

SUMMARY

(2) Advantages of the prior art can be overcome and benefits as described later in this disclosure can be achieved through the provision of systems and related temperature calibration methods. Various implementations of the apparatus and methods are described below, and the apparatus and methods, including and excluding the additional implementations enumerated below, in any

combination (provided these combinations are not inconsistent), may overcome these shortcomings and achieve the benefits described herein.

(3) In accordance with a first implementation, an apparatus includes a flow cell interface, a temperature control device, an infrared sensor, and a controller. The flow cell interface includes a flow cell support and the temperature control device is for the flow cell support. The controller is to command the temperature control device to cause the flow cell support to achieve a temperature value, cause the infrared sensor to measure an actual temperature value of the flow cell support, and calibrate the temperature control device based on a difference between the commanded temperature value and the actual temperature value.

(4) In accordance with a second implementation, a method includes commanding a temperature control device of a system to cause a flow cell support to achieve a temperature value. The method also includes measuring an actual temperature value of the flow cell support using an infrared sensor and calibrating the temperature control device based on a difference between the commanded temperature value and the actual temperature value.

(5) In accordance with a third implementation, a method includes commanding a temperature control device of a system to cause a flow cell support to achieve a temperature value and measuring a plurality of actual temperature value of the flow cell support using an infrared sensor. The method also includes determining an average actual temperature value based on the plurality of actual temperature values and calibrating the temperature control device based on a difference between the commanded temperature value and the average actual temperature value.

(6) In accordance with a fourth implementation, an apparatus includes a flow cell interface, a temperature control device, an infrared sensor, and a controller. The flow cell interface includes a flow cell support and the temperature control device is for the flow cell support. The controller is to command the temperature control device to cause the flow cell support to achieve a temperature value, and cause the infrared sensor to measure a plurality of actual temperature values of the flow cell support.

(7) In accordance with a fifth implementation, a method includes commanding a temperature control device to cause a flow cell support to achieve a temperature value; measuring an actual temperature value of a flow cell disposed on the flow cell support using an infrared sensor; comparing the commanded temperature value and the actual temperature value; and generating an alert when a difference between the commanded temperature value and the actual temperature value is greater than a threshold.

(8) In accordance with a sixth implementation, an apparatus includes a flow cell interface, a temperature control device, an infrared sensor, a temperature sensor, and a controller. The flow cell interface includes a flow cell support and the temperature control device is for the flow cell support. The infrared sensor is to measure a first temperature value adjacent to the temperature sensor; the temperature sensor is to measure a second temperature value; and the controller is to calibrate the infrared sensor based on a difference between the first temperature value and the second temperature value.

(9) In accordance with a seventh implementation, an apparatus includes a flow cell interface, a temperature control device, an infrared sensor, and a controller. The flow cell interface including a flow cell support and the temperature control device for the flow cell support. The controller to command the temperature control device to cause the flow cell support to achieve a temperature value, and cause the infrared sensor to measure a plurality of actual temperature values of the flow cell support.

(10) In further accordance with the foregoing first, second, third, fourth, fifth, sixth, and/or seventh implementations, an apparatus and/or method may further comprise or include any one or more of the following:

(11) In accordance with an implementation, the temperature control device includes a thermoelectric cooler.

- (12) In accordance with another implementation, the temperature control device includes a resistance temperature detector.
- (13) In accordance with another implementation, the controller is to calibrate the temperature control device by calibrating the resistance temperature detector.
- (14) In accordance with another implementation, the apparatus includes an imaging system. The infrared sensor is coupled to the imaging system.
- (15) In accordance with another implementation, the apparatus includes a bracket coupling the infrared sensor to the imaging system.
- (16) In accordance with another implementation, the controller is further to calibrate the infrared sensor.
- (17) In accordance with another implementation, the apparatus also includes a temperature sensor. The infrared sensor is to measure a first temperature value adjacent to the temperature sensor, the temperature sensor is to measure a second temperature value, and the controller is to calibrate the infrared sensor based on a difference between the first temperature value and the second temperature value.
- (18) In accordance with another implementation, the flow cell interface further includes a flow cell deck carrying the flow cell support. The temperature sensor is coupled to the flow cell deck.
- (19) In accordance with another implementation, the apparatus includes a printed circuit board coupled to the temperature sensor. The printed circuit board is positioned between the temperature sensor and the flow cell deck.
- (20) In accordance with another implementation, the apparatus includes a metallic layer covering a surface of the temperature sensor.
- (21) In accordance with another implementation, the metallic layer and a surface of the flow cell support have substantially similar emissivity.
- (22) In accordance with another implementation, the temperature sensor and the metallic layer are spaced from the flow cell support.
- (23) In accordance with another implementation, the metallic layer includes aluminum.
- (24) In accordance with another implementation, the apparatus includes thermally conductive epoxy coupling the temperature sensor and the metallic layer.
- (25) In accordance with another implementation, the apparatus also includes a temperature sensor. The infrared sensor is to measure a first temperature value of the flow cell support, the temperature sensor is to measure a second temperature value at the flow cell support, and the controller is to calibrate the infrared sensor based on a difference between the first temperature value and the second temperature value.
- (26) In accordance with another implementation, the temperature sensor is coupled to the flow cell support.
- (27) In accordance with another implementation, the apparatus also includes a printed circuit board coupled to the temperature sensor. The temperature sensor is positioned between the flow cell interface and the printed circuit board.
- (28) In accordance with another implementation, a surface of the flow cell support includes a metallic layer.
- (29) In accordance with another implementation, the temperature sensor includes a digital temperature sensor.
- (30) In accordance with another implementation, commanding the temperature control device to cause the flow cell support to achieve the temperature value includes: commanding a thermoelectric cooler to cause the flow cell support to achieve the temperature value; and measuring the temperature value with a resistance temperature detector.
- (31) In accordance with another implementation, measuring the actual temperature value for the flow cell support using the infrared sensor includes measuring the actual temperature value for the flow cell support using the infrared sensor while moving the flow cell interface relative to the

infrared sensor.

(32) In accordance with another implementation, the method also includes commanding the temperature control device to cause the flow cell support to achieve a second temperature value, measuring a second actual temperature value of a flow cell disposed on the flow cell support using the infrared sensor, comparing the second commanded temperature value and the second actual temperature value, and recalibrating the temperature control device based the comparison.

(33) In accordance with another implementation, the method also includes: causing the system to perform at least a portion of an analysis on one or more samples of interest within a flow cell disposed on the flow cell support, commanding the temperature control device to cause the flow cell support to achieve a second temperature value; measuring a second actual temperature value of the flow cell disposed on the flow cell support using the infrared sensor; comparing the second commanded temperature value and the second actual temperature value; and recalibrating the temperature control device based the comparison.

(34) In accordance with another implementation, calibrating the temperature control device is responsive to receiving a command at the controller to calibrate the temperature control device.

(35) In accordance with another implementation, receiving the command at the controller to calibrate the temperature control device includes receiving the command from a remote computing device.

(36) In accordance with another implementation, receiving the command at the controller to calibrate the temperature control device includes receiving user input at a user interface of the system.

(37) In accordance with another implementation, the method includes calibrating the infrared sensor.

(38) In accordance with another implementation, calibrating the infrared sensor includes: measuring a first temperature value adjacent to a temperature sensor carried by the flow cell interface using the infrared sensor, measuring a second temperature value using the temperature sensor, and calibrating the infrared sensor with a controller based on a difference between the first temperature value and the second temperature value.

(39) In accordance with another implementation, calibrating the infrared sensor includes: measuring a first temperature value of the flow cell support using the infrared sensor, measuring a second temperature value at the flow cell support using a temperature sensor where the temperature sensor is carried by the flow cell support, and calibrating the infrared sensor with the controller based on a difference between the first temperature value and the second temperature value.

(40) In accordance with another implementation, calibrating the infrared sensor is responsive to receiving a command at the controller to calibrate the infrared sensor.

(41) In accordance with another implementation, receiving the command at the controller to calibrate the infrared sensor includes receiving the command from a remote computing device.

(42) In accordance with another implementation, receiving the command at the controller to calibrate the infrared sensor includes receiving user input at a user interface of the system.

(43) In accordance with another implementation, calibrating the infrared sensor includes generating data associated with the difference between the first temperature value and the second temperature value and calibrating the infrared sensor with the controller based on the data.

(44) In accordance with another implementation, commanding the temperature control device of the system to cause the flow cell support to achieve the temperature value includes commanding a first zone of the temperature control device of the system to cause a first zone of the flow cell support to achieve the temperature value and wherein measuring the actual temperature value of the flow cell support using the infrared sensor includes measuring an actual temperature value of the first zone of the flow cell support.

(45) In accordance with another implementation, calibrating the temperature control device based on a difference between the commanded temperature value and the actual temperature value

includes calibrating a resistance temperature detector of the first zone of the temperature control device based on a difference between the commanded temperature value and the actual temperature value.

(46) In accordance with another implementation, the method also includes 1) commanding a second zone of the temperature control device of the system to cause a second zone of the flow cell support to achieve a second temperature value; 2) measuring an actual temperature value of the second zone of flow cell support using the infrared sensor; and 3) calibrating a resistance temperature detector of the second zone of the temperature control device based on a difference between the commanded temperature value and the actual temperature value.

(47) In accordance with another implementation, the controller is to generate a heat map based on the plurality of actual temperature values measured.

(48) In accordance with another implementation, the controller is to diagnosis an error based on the heat map.

(49) In accordance with another implementation, the controller is to calibrate the temperature control device based on the heat map.

(50) In accordance with another implementation, the heat map includes a 2-D heat map.

(51) In accordance with another implementation, the method also includes calibrating the temperature control device based on the difference between the commanded temperature value and the actual temperature value.

(52) In accordance with another implementation, the apparatus includes a flow cell carrying the temperature sensor.

(53) In accordance with another implementation, the flow cell includes a mock flow cell.

(54) It should be appreciated that all combinations of the foregoing concepts and additional concepts discussed in greater detail below (provided such concepts are not mutually inconsistent) are contemplated as being part of the subject matter disclosed herein and/or may be combined to achieve the particular benefits of a particular aspect. In particular, all combinations of claimed subject matter appearing at the end of this disclosure are contemplated as being part of the subject matter disclosed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a schematic diagram of an implementation of a system in accordance with the teachings of this disclosure.
- (2) FIG. 2 is an isometric view a flow cell interface that can be used to implement the flow cell interface of FIG. 1 and an isometric view of the imaging system of FIG. 1.
- (3) FIG. 3 is a cross-sectional schematic view of a portion of a flow cell interface of FIG. 2 showing the printed circuit board positioned between the temperature sensor and the flow cell interface.
- (4) FIG. 4 is an isometric view of another flow cell interface that can be used to implement the flow cell interface of FIG. 1 and an isometric schematic view of the imaging system of FIG. 1.
- (5) FIG. 5 is a cross-sectional schematic view of a portion of the flow cell interface of FIG. 4 showing a temperature sensor coupled to a lower surface of a flow cell support of the flow cell interface.
- (6) FIG. 6 is a top plan view of a temperature control device that can be used to implement the temperature control device of FIG. 1.
- (7) FIG. 7 illustrates a flowchart for a process of calibrating the temperature control devices of FIGS. 1, 2, 4, and 6 or any of the other implementations disclosed herein.
- (8) FIG. 8 illustrates another flowchart for a process of calibrating the temperature control devices

of FIGS. 1, 2, 4, and 6 or any of the other implementations disclosed herein.

(9) FIG. 9 illustrates another flowchart for a process of calibrating the temperature control devices of FIGS. 1, 2, 4, and 6 or any of the other implementations disclosed herein.

(10) FIG. 10 illustrates a flowchart for a process of calibrating the temperature control devices of FIGS. 1, 2, 4, and 6 and the infrared sensors of FIGS. 1, 2, and 4 or any of the other implementations disclosed herein.

DETAILED DESCRIPTION

(11) Although the following text discloses a detailed description of implementations of methods, apparatuses and/or articles of manufacture, it should be understood that the legal scope of the property right is defined by the words of the claims set forth at the end of this patent. Accordingly, the following detailed description is to be construed as examples only and does not describe every possible implementation, as describing every possible implementation would be impractical, if not impossible. Numerous alternative implementations could be implemented, using either current technology or technology developed after the filing date of this patent. It is envisioned that such alternative implementations would still fall within the scope of the claims.

(12) The implementations disclosed herein relate to thermal calibration of heating/cooling components and/or temperature sensing components used in sequencing or other systems. Thermal calibration allows for more accurate sequencing performance, as enzymes and sequencing chemistry used in sequencing can be temperature sensitive, and thus thermal calibration is more accurate. The systems include a flow cell support, a temperature control device, an infrared sensor oriented to measure a temperature of the flow cell support, and a controller. The controller calibrates the temperature control device by determining a difference between the temperature measured by the infrared sensor and the temperature that the temperature control device was commanded to generate.

(13) The infrared sensor can be calibrated through the use of a temperature sensor coupled to a structure of the system in some implementations. The infrared sensor measures a first temperature value at or adjacent to the temperature sensor and the temperature sensor measures a second temperature value. The controller of the system calibrates the infrared sensor in some implementations by determining a difference between the temperature measured by the infrared sensor and the temperature measured by the temperature sensor.

(14) The disclosed calibration processes allow the temperature control devices to be accurately calibrated without the use of costly calibration tools and/or without a service technician being on site with the system. The disclosed calibration processes also allow for frequent calibrations and fast remote detection and diagnosis of errors. Higher quality temperature calibration can be achieved as a result leading to less downtime, higher sequencing data quality, and/or less spatial variation in data over the flow cell and between flow cells

(15) FIG. 1 illustrates a schematic diagram of an implementation of a system **100** in accordance with the teachings of this disclosure. The system **100** can be used to perform an analysis on one or more samples of interest. The sample may include one or more DNA clusters that have been linearized to form a single stranded DNA (sstDNA). The system **100** includes a flow cell interface **102** having a flow cell support **104** that is adapted to support a flow cell assembly **106** including a corresponding flow cell **108** in the implementation shown. The flow cell interface **102** may be associated with and/or referred to as a flow cell deck and the flow cell support **104** may be associated with and/or referred to as a flow cell chuck. The flow cell support **104** can include a vacuum channel, latches, a snap fit mechanism, and/or a tongue-and-groove coupling that is used to secure the flow cell assembly **106** to the flow cell support **104**.

(16) The system **100** also includes, in part, a temperature control device **110** for the flow cell support **104**, an infrared sensor **112**, an imaging system **114**, a stage assembly **116**, a reagent selector valve assembly **118** that has a reagent selector valve **120** and a valve drive assembly **122**, and a controller **124**. The reagent selector valve assembly **118** may be referred to as a mini-valve

assembly. The controller **124** is electrically and/or communicatively coupled to components of the system **100**, such as the temperature control device **110**, the infrared sensor **112**, the imaging system **114**, the stage assembly **116**, and the reagent selector valve assembly **118** and is adapted to cause the temperature control device **110**, the infrared sensor **112**, the imaging system **114**, the stage assembly **116**, and the reagent selector valve assembly **118** to perform various functions as disclosed herein.

(17) The controller **124** in operation commands the temperature control device **110** to cause the flow cell support **104** to achieve a temperature value and causes the infrared sensor **112** to measure an actual temperature value of the flow cell support **104**. The controller **124** may additionally or alternatively cause the infrared sensor **112** to measure an actual temperature value of the flow cell **108** disposed on the flow cell support **104**. The system **100** may determine the thermal health of sequencing chemistry operations by determining the actual temperature of the flow cell **108** as an example.

(18) The controller **124** calibrates the temperature control device **110** in this way based on a difference between the commanded temperature value and the actual temperature value. The system **100** can perform the calibration procedure in a self-diagnostic/self-calibration mode and/or a service technician can communicate with the system **100** to perform maintenance on the system **100** and/or calibrate the temperature control device **110**. The service technician may calibrate the system **100** in the same location as the system **100** or in a location remote to the system **100** by remotely accessing and/or communicating with the system **100**. The system **100** can perform the calibration procedure on a schedule such as before an analysis takes place, while an analysis is taking place, and/or after an analysis takes place, for example. The calibration procedure can be performed at any time, however. The controller **124** may additionally or alternatively generate an alert when a difference between the commanded temperature value and the actual temperature value is greater than a threshold and/or calibrate the temperature control device **110** based on the difference between the commanded temperature value and the actual temperature value. More generally, the controller **124** can perform diagnostics based on the actual temperature values.

(19) In instances when the controller **124** causes the infrared sensor **112** to measure a plurality of actual temperature values of the flow cell support **104**, the controller **124** may generate a heat map based on the plurality of actual temperature values measured. The controller **124** can diagnosis an error based on the heat map and/or calibrate the temperature control device **110** based on the heat map in some implementations. The heat map may be a 2-D heat map.

(20) The controller **124** can initiate the calibration procedure in response to receiving a command. The controller **124** can receive the command based on the system **100** initiating and/or performing a process and/or based on a schedule. The controller **124** can receive the command from a remote computing device and/or from an individual using the system **100**, however. The remote computing device can be used to calibrate the system **100** to ensure that the system **100** is working properly and/or is within a threshold operating range.

(21) The system **100** also allows the remote computing device to perform tests and/or to diagnose potential errors in the system **100**. Put another way, the system **100** allows a service technician to perform maintenance on the system **100** and to identify potential issues with components of the system **100** such as the temperature control device **110**, the infrared sensor **112**, and/or the flow cell interface **102**. The controller **124** can automatically report or output any identified issues or other data to inform the service technician of needed service. While this example mentions a remote computing device accessing the system **100** to perform maintenance and/or diagnostics, the system **100** may also perform a “self-calibration” where the system **100** calibrates the temperature control device **110** and/or the infrared sensor **112**. The system **100** can, thus, perform the calibration processes and/or diagnostic processes disclosed with or without input from a service technician.

(22) The controller **124** can calibrate the temperature control device **110** and/or perform diagnostics prior to the system **100** performing an analysis and the flow cell assembly **106** may or may not be

placed on the flow cell support **104** during the calibration process. The infrared sensor **112** measures a temperature of the flow cell **108** on the flow cell support **104** in some implementations and, thus, does not directly measure the temperature of the flow cell support **104**. The controller **124** can account for a difference between the temperature determined by the resistance temperature detectors **128** at the temperature control device **110** and the temperature determined by the thermoelectric device **110** at the flow cell **108**. The system **100** can also perform the calibration procedure and/or perform diagnostics after a portion of an analysis has taken place of one or samples of interest contained within the flow cell **108**.

(23) The stage assembly **116** can move the flow cell interface **102** and/or the flow cell support **104** relative to the infrared sensor **112** during the calibration process regardless of the presence of the flow cell **108** to allow the infrared sensor **112** to obtain temperature data for a plurality of locations on the flow cell support **104** and/or on the flow cell **108**. The controller **124** can use the measured temperature values to determine a temperature across the flow cell support **104** or the flow cell **108** and/or to determine if a temperature gradient across the flow cell support **104** or the flow cell **108** is within an acceptable threshold, for example.

(24) The temperature control device **110** is shown including one or more thermoelectric coolers **126** and including one or more resistance temperature detectors **128**. The thermoelectric coolers **126** may be Peltier devices. While resistance temperature detectors **128** are mentioned being included, the temperature control device **110** may alternatively include any other type of thermometer.

(25) The temperature control device **110** may be positioned underneath the flow cell support **104** and heats/cools the flow cell support **104** and, thus, heats/cools the flow cell **108** on the flow cell support **104**. The temperature control device **110** can include different zones **502**, **504**, **506** (see, FIG. 5) and the controller **124** can command the thermoelectric coolers **126** to achieve a temperature at one or more of the zones **502**, **504**, **506**. The corresponding resistance temperature detector **128** can determine an actual temperature of the zone **502**, **504**, **506** and provide feedback to the controller **124** that allows the thermoelectric cooler **126** to achieve the commanded temperature value. A feedback loop is thus provided between the controller **124** and the resistance temperature detectors **128**. The controller **124** can calibrate the temperature control device **110** by calibrating the resistance temperature detectors **128** of the temperature control device **110** in some implementations.

(26) The infrared sensor **112** can obtain a plurality of actual temperature values of the flow cell support **104** and the controller **124** can determine an average actual temperature value based on the plurality of actual temperature values. The controller **124** can calibrate the temperature control device **110** based on a difference between the commanded temperature value and the average actual temperature value. The infrared sensor **112** can also or alternatively determine the actual temperature value for a plurality of locations on one or more of the zones **502**, **504**, **506** of the temperature control device **110** and the controller **124** can determine an average actual temperature value for the corresponding zones **502**, **504**, **506** based on the actual temperature values. The controller **124** can calibrate the resistance temperature detectors **128** for the corresponding zones **502**, **504**, **506** based on a difference between the commanded temperature values and the average actual temperature values. The controller **124** may alternatively calibrate the resistance temperature detectors **128** for the corresponding zones **502**, **504**, **506** based on one or more actual temperature values from the infrared sensor **112**. The infrared sensor **112** can also or alternatively determine the actual temperature value for a plurality of locations on one or more of the zones **502**, **504**, **506** of the temperature control device **110** and the controller **124** can generate a heat map and/or perform diagnostics based on the actual temperature values.

(27) The process of the controller **124** calibrating the temperature control device **110** can include the controller **124** calibrating the resistance temperature detectors **128**. The controller **124** can calibrate the resistance temperature detectors **128** to more accurately measure the temperature generated by the corresponding thermoelectric cooler **126**. Put another way, the controller **124** can

calibrate the resistance temperature detectors **128** to compensate for deviation between a measured temperature value and an actual temperature value. The temperature control device **110** includes two thermoelectric coolers **126** and two resistance temperature detectors **128** in the implementation shown. The thermoelectric cooler **126** can, however, include any number of thermoelectric coolers **126** and/or any number of resistance temperature detectors **128**.

(28) The controller **124** can calibrate the resistance temperature detectors **128** at any time so that the resistance temperature detectors **128** more accurately measure the temperature generated by the corresponding thermoelectric coolers **126** that is delivered to the flow cell **108**. The controller **124** can offset the measurements of the resistance temperature detectors **128** or revise settings of the resistance temperature detectors **128** for calibration.

(29) The infrared sensor **112** is coupled to the imaging system **114**, in the implementation shown. The infrared sensor **112** is positioned to allow the infrared sensor **112** to measure temperature values of the flow cell support **104** and/or the flow cell interface **102**. The stage assembly **116** can move the flow cell interface **102** and the flow cell support **104** relative to the infrared sensor **112** to allow the temperature values to be obtained.

(30) A bracket **130** is shown coupling the infrared sensor **112** and the imaging system **114**. The bracket **130** may have apertures that allow the infrared sensor **112** to be attached to the bracket **130**. The bracket **130** may also or alternatively be an L-shaped bracket. Other ways of coupling the infrared sensor **112** and the imaging system **114** may prove suitable, however. The infrared sensor **112** may alternatively be coupled in a different location.

(31) The controller **124** can also calibrate the infrared sensor **112**. The system **100** may calibrate the infrared sensor **112** prior to calibrating the temperature control device **110**, on a different schedule, and/or at any time. The system **100** includes a temperature sensor **132** to do so. The temperature sensor **132** may be a digital temperature sensor such as a ADT7422. Other temperature sensors **132** may prove suitable. The flow cell **108** may alternatively carry the temperature sensor **132** in other implementations. The flow cell **108** may be considered a mock flow cell in implementations when the flow cell **108** carries the temperature sensor **132**. The mock flow cell may be used for testing and/or calibration purposes as an example.

(32) The flow cell interface **102** includes a flow cell deck **134** that carries the flow cell support **104** and to which the temperature sensor **132** is coupled in the implementation shown. The temperature sensor **132** can be coupled in different locations, however.

(33) The infrared sensor **112** measures a first temperature value adjacent to the temperature sensor **132**, the temperature sensor **132** measures a second temperature value, and the controller **124** calibrates the infrared sensor **112** based on a difference between the first temperature value and the second temperature value. The controller **124** can calibrate the infrared sensor **112** by offsetting the measurements of the infrared sensor **112** or revising the settings of the infrared sensor **112**.

(34) The temperature sensor **132** may be a part of a calibration assembly **135** that includes the temperature sensor **132** and a printed circuit board **136**. The printed circuit board **136** is shown coupled to the temperature sensor **132** and positioned between the temperature sensor **132** and the flow cell deck **134**. The temperature sensor **132** and/or the printed circuit board **136** may be in another position, however. The temperature sensor **132** may be coupled to the flow cell support **104** and the printed circuit board **136** such that the temperature sensor **132** is positioned between the flow cell support **104** and the printed circuit board **136**, for example (see, for example, FIG. 5).

(35) The calibration assembly **135** may also include a metallic layer **138**. The metallic layer **138** is shown covering a surface **140** of the temperature sensor **132**. The infrared sensor **112** can, thus, measure the first temperature value at the metallic layer **138** when the infrared sensor **112** is being calibrated. The metallic layer **138** may include aluminum. The metallic layer **138** can include additional or alternative materials, however. Thermally conductive epoxy **142** couples the temperature sensor **132** and the metallic layer **138**. The temperature sensor **132** can measure the second temperature value at the metallic layer **138** relatively accurately as a result of the presence

of the thermally conductive epoxy **142**.

(36) The metallic layer **138** and a surface **139** of the flow cell support **104** can have substantially similar emissivity such as having similar surface finishes. The surface **139** of the flow cell support **104** can include a metallic layer.

(37) The stage assembly **116** includes an x-stage **144** and a y-stage **146**. The x-stage **144** moves the flow cell interface **102** in an x-direction relative to the imaging system **114** and the infrared sensor **112** and the y-stage **146** moves the flow cell interface **102** in the y-direction relative to the imaging system **114** and the infrared sensor **112**. The x-stage **144** and/or the y-stage **146** may be linear stages. The x-stage **144** and/or the y-stage **146** may be any other type of motor or actuator, however. The stage assembly **116** can move the flow cell support **104** to different locations to allow the infrared sensor **112** to measure one or more temperature values of the flow cell support **104**, for example.

(38) Referring still to the system **100** of FIG. **1**, the system **100** also includes a sipper manifold assembly **148**, a sample loading manifold assembly **150**, a pump manifold assembly **152**, a drive assembly **154**, and a waste reservoir **156** in the implementation shown. The controller **124** is electrically and/or communicatively coupled to the sipper manifold assembly **148**, the sample loading manifold assembly **150**, the pump manifold assembly **152**, and the drive assembly **154** and is adapted to cause the sipper manifold assembly **148**, the sample manifold assembly **150**, the pump manifold assembly **152**, and the drive assembly **154** to perform various functions as disclosed herein.

(39) Referring to the flow cell **108**, each of the flow cells **108** includes a plurality of channels **158** in the implementation shown, each having a first channel opening positioned at a first end of the flow cell **108** and a second channel opening positioned at a second end of the flow cell **108**. Depending on the direction of flow through the channels **158**, either of the channel openings may act as an inlet or an outlet. While the flow cell **108** is shown including two channels **158** in FIG. **1**, any number of channels **158** may be included (e.g., 1, 2, 6, 8).

(40) The flow cell assembly **106** also includes a flow cell frame **160** and a flow cell manifold **162** coupled to the first end of the corresponding flow cell **108**. As used herein, a “flow cell” (also referred to as a flowcell) can include a device having a lid extending over a reaction structure to form a flow channel therebetween that is in communication with a plurality of reaction sites of the reaction structure. Some flow cells may also include a detection device that detects designated reactions that occur at or proximate to the reaction sites. As shown, the flow cell **108**, the flow cell manifold **162**, and/or any associated gaskets used to establish a fluidic connection between the flow cell **108** and the system **100** are coupled or otherwise carried by the flow cell frame **160**. While the flow cell frame **160** is shown included with the flow cell assembly **106** of FIG. **1**, the flow cell frame **160** may be omitted. As such, the flow cell **108** and the associated flow cell manifold **162** and/or gaskets may be used with the system **100** without the flow cell frame **160**.

(41) Prior to referring to some of the additional components of the system **100** of FIG. **1** such as some of the fluidic components, it is noted that while some components of the system **100** are shown once and coupled to the single flow cell **108**, in some implementations, these components may be duplicated, thereby allowing more flow cells **108** to be used with the system **100** (e.g., 2, 3, 4) and each flow cell **108** can have its own corresponding components as a result. Each flow cell **108** may be associated with a separate sample cartridge **164**, sample loading manifold assembly **150**, pump manifold assembly **152**, etc. when more than one flow cell **108** is included with the system **100**.

(42) Referring now to the sample cartridge **164**, the sample loading manifold assembly **150**, and the pump manifold assembly **152**, the system **100** includes a sample cartridge receptacle **166** that receives the sample cartridge **164** that carries one or more samples of interest (e.g., an analyte) in the implementation shown. The system **100** also includes a sample cartridge interface **168** that establishes a fluidic connection with the sample cartridge **164**.

(43) The sample loading manifold assembly **150** includes one or more sample valves **170** and the pump manifold assembly **152** includes one or more pumps **172**, one or more pump valves **174**, and a cache **176**. One or more of the valves **170**, **174** may be implemented by a rotary valve, a pinch valve, a flat valve, a solenoid valve, a check valve, a piezo valve, and/or a three-way valve. However, different types of fluid control devices may be used. One or more of the pumps **172** may be implemented by a syringe pump, a peristaltic pump, and/or a diaphragm pump. Other types of fluid transfer devices may be used, however. The cache **176** may be a serpentine cache and may temporarily store one or more reaction components during bypass manipulations of the system **100** of FIG. **1**. While the cache **176** is shown being included in the pump manifold assembly **152**, in another implementation, the cache **176** may be located in a different location. In certain implementations, the cache **176** may be included in the sipper manifold assembly **148** or in another manifold downstream of a bypass fluidic line **178**.

(44) The sample loading manifold assembly **150** and the pump manifold assembly **152** flow one or more samples of interest from the sample cartridge **164** through a fluidic line **180** toward the flow cell assembly **106**. In some implementations, the sample loading manifold assembly **150** can individually load/address each channel **158** of the flow cell **108** with a sample of interest. The process of loading the channels **158** of the flow cell **108** with a sample of interest may occur automatically using the system **100** of FIG. **1**.

(45) The sample cartridge **164** and the sample loading manifold assembly **150** are positioned downstream of the flow cell assembly **106** in the system **100** of FIG. **1**. The sample loading manifold assembly **150** may load a sample of interest into the flow cell **108** from the rear of the flow cell **108**. Loading a sample of interest from the rear of the flow cell **108** may be referred to as “back loading.” Back loading the sample of interest into the flow cell **108** may reduce contamination. The sample loading manifold assembly **50** is coupled between the flow cell assembly **106** and the pump manifold assembly **52**.

(46) To draw a sample of interest from the sample cartridge **164** and toward the pump manifold assembly **152**, the sample valves **170**, the pump valves **174**, and/or the pumps **172** may be selectively actuated to urge the sample of interest toward the pump manifold assembly **152**. The sample cartridge **164** may include a plurality of sample reservoirs that are selectively fluidically accessible via the corresponding sample valve **170**. Each sample reservoir can thus be selectively isolated from other sample reservoirs using the corresponding sample valves **170**.

(47) To individually flow the sample of interest toward a corresponding channel of one of the flow cells **108** and away from the pump manifold assembly **152**, the sample valves **170**, the pump valves **174**, and/or the pumps **172** can be selectively actuated to urge the sample of interest toward the flow cell assembly **106** and into the respective channels **58** of the corresponding flow cell **108**. Each channel **158** of the flow cell **108** receives the sample of interest in some implementations. In other implementations, one or more of the channels **158** of the flow cell(s) **108** selectively receives the sample of interest and others of the channels **158** of the flow cell(s) **108** do not receive the sample of interest. The channels **158** of the flow cell(s) **108** that may not receive the sample of interest may receive a wash buffer instead.

(48) The drive assembly **154** interfaces with the sipper manifold assembly **148** and the pump manifold assembly **152** to flow one or more reagents that interact with the sample within the corresponding flow cell **108**. In an implementation, a reversible terminator is attached to the reagent to allow a single nucleotide to be incorporated onto a growing DNA strand. One or more of the nucleotides has a unique fluorescent label that emits a color when excited in some such implementations. The color (or absence thereof) is used to detect the corresponding nucleotide. The imaging system **114** excites one or more of the identifiable labels (e.g., a fluorescent label) in the implementation shown and thereafter obtains image data for the identifiable labels. The labels may be excited by incident light and/or a laser and the image data may include one or more colors emitted by the respective labels in response to the excitation. The image data (e.g., detection data)

may be analyzed by the system **100**. The imaging system **114** may be a fluorescence spectrophotometer including an objective lens and/or a solid-state imaging device. The solid-state imaging device may include a charge coupled device (CCD) and/or a complementary metal oxide semiconductor (CMOS). Other types of imaging systems and/or optical instruments may be used, however. The imaging system **114** may be or may be associated with a scanning electron microscope, a transmission electron microscope, an imaging flow cytometer, high-resolution optical microscopy, confocal microscopy, epifluorescence microscopy, two photon microscopy, differential interference contrast microscopy, etc. in certain implementations.

(49) After the image data is obtained, the drive assembly **154** interfaces with the sipper manifold assembly **148** and the pump manifold assembly **152** to flow another reaction component (e.g., a reagent) through the flow cell **108** that is thereafter received by the waste reservoir **156** via a primary waste fluidic line **182** and/or otherwise exhausted by the system **100**. Some reaction components perform a flushing operation that chemically cleaves the fluorescent label and the reversible terminator from the sstDNA. The sstDNA is then ready for another cycle.

(50) The primary waste fluidic line **182** is coupled between the pump manifold assembly **152** and the waste reservoir **156**. The pumps **172** and/or the pump valves **174** of the pump manifold assembly **152** selectively flow the reaction components from the flow cell assembly **106**, through the fluidic line **180** and the sample loading manifold assembly **150** to the primary waste fluidic line **182** in some implementations.

(51) The flow cell assembly **106** is coupled to a central valve **184** via the flow cell interface **102**. An auxiliary waste fluidic line **186** is coupled to the central valve **184** and to the waste reservoir **156**. The auxiliary waste fluidic line **186** receives excess fluid of a sample of interest from the flow cell assembly **106** in some implementations, via the central valve **184**, and flows the excess fluid of the sample of interest to the waste reservoir **156** when back loading the sample of interest into the flow cell **108**, as described herein. That is, the sample of interest may be loaded from the rear of the flow cell **108** and any excess fluid for the sample of interest may exit from the front of the flow cell **108**. By back loading samples of interest into the flow cell **108**, different samples can be separately loaded to corresponding channels **158** of the corresponding flow cell **108** and the single flow cell manifold **162** can couple the front of the flow cell **108** to the central valve **184** to direct excess fluid of each sample of interest to the auxiliary waste fluidic line **186**. Once the samples of interest are loaded into the flow cell **108**, the flow cell manifold **162** can be used to deliver common reagents from the front of the flow cell **108** (e.g., upstream) for each channel **158** of the flow cell **108** that exit from the rear of the flow cell **108** (e.g., downstream). Put another way, the sample of interest and the reagents may flow in opposite directions through the channels **158** of the flow cell **108**.

(52) Referring to the sipper manifold assembly **148**, the sipper manifold assembly **148** includes a shared line valve **188** and a bypass valve **190** in the implementation shown. The shared line valve **188** may be referred to as a reagent selector valve. The valve **120** of the reagent selector valve assembly **118**, the central valve **184**, and/or the valves **188**, **190** of the sipper manifold assembly **148** may be selectively actuated to control the flow of fluid through fluidic lines **178**, **192**, **194**, **196**, **198**. One or more of the valves **120**, **170**, **174**, **184**, **188** may be implemented by a rotary valve, a pinch valve, a flat valve, a solenoid valve, a check valve, a piezo valve, and/or a three-way valve. Other fluid control devices may prove suitable.

(53) The sipper manifold assembly **148** may be coupled to a corresponding number of reagents reservoirs **200** via reagent sippers **202**. The reagent reservoirs **200** may contain fluid (e.g., reagent and/or another reaction component). The sipper manifold assembly **148** includes a plurality of ports in some implementations. Each port of the sipper manifold assembly **148** may receive one of the reagent sippers **202**. The reagent sippers **202** may be referred to as fluidic lines. While the system **100** includes the sipper manifold assembly **148**, the system **100** may alternatively receive a reagent cartridge and, thus, the sipper manifold assembly **148** may be modified to omit the reagent sippers **202** and/or to include an alternative fluidic interface, for example, or the sipper manifold assembly

148 may be omitted.

(54) The shared line valve **188** of the sipper manifold assembly **148** is coupled to the central valve **184** via the shared reagent fluidic line **192**. Different reagents may flow through the shared reagent fluidic line **192** at different times. In an implementation, the pump manifold assembly **152** may draw wash buffer through the shared reagent fluidic line **192**, the central valve **184**, and the corresponding flow cell assembly **106** when performing a flushing operation before changing between one reagent and another. The shared reagent fluidic line **192** may, thus, be involved in the flushing operation. While one shared reagent fluidic line **192** is shown, any number of shared fluidic lines may be included in the system **100**.

(55) The bypass valve **190** of the sipper manifold assembly **148** is coupled to the central valve **184** via the reagent fluidic lines **194**, **196**. The central valve **184** may have one or more ports that correspond to the reagent fluidic lines **194**, **196**.

(56) The dedicated fluidic lines **198** are coupled between the sipper manifold assembly **148** and the reagent selector valve assembly **118**. Each of the dedicated reagent fluidic lines **198** may be associated with a single reagent. The fluids that flow through the dedicated reagent fluidic lines **198** may be used during sequencing operations and may include a cleave reagent, an incorporation reagent, a scan reagent, a cleave wash, and/or a wash buffer. Because only a single reagent may flow through each of the dedicated reagent fluidic lines **198**, the dedicated reagent fluidic lines **198** themselves may not be flushed when performing a flushing operation before changing between one reagent and another. The approach of including dedicated reagent fluidic lines **198** may be helpful when the system **100** uses reagents that may have adverse reactions with other reagents. Reducing a number of fluidic lines or a length of the fluidic lines that are flushed when changing between different reagents moreover reduces reagent consumption and flush volume and may decrease cycle times of the system **100**. While two dedicated reagent fluidic lines **198** are shown, any number of dedicated fluidic lines may be included in the system **100**.

(57) The bypass valve **190** is also coupled to the cache **176** of the pump manifold assembly **152** via the bypass fluidic line **178**. One or more reagent priming operations, hydration operations, mixing operations, and/or transfer operations may be performed using the bypass fluidic line **178**. The priming operations, the hydration operations, the mixing operations, and/or the transfer operations may be performed independent of the flow cell assembly **106**. The operations using the bypass fluidic line **178** may thus occur during incubation of one or more samples of interest within the flow cell assembly **106**. That is, the shared line valve **188** can be utilized independently of the bypass valve **190** such that the bypass valve **190** can utilize the bypass fluidic line **178** and/or the cache **176** to perform one or more operations while the shared line valve **188** and/or the central valve **184** simultaneously, substantially simultaneously, or offset synchronously perform other operations. The system **100** can thus perform multiple operations at once, thereby reducing run time.

(58) Referring now to the drive assembly **154**, in the implementation shown, the drive assembly **154** includes a pump drive assembly **204** and a valve drive assembly **206**. The pump drive assembly **204** may be adapted to interface with the one or more pumps **172** to pump fluid through the flow cell **108** and/or to load one or more samples of interest into the flow cell **108**. The valve drive assembly **206** may be adapted to interface with one or more of the valves **120**, **170**, **174**, **184**, **188**, **190** to control the position of the corresponding valves **120**, **170**, **174**, **184**, **188**, **190**.

(59) Referring to the controller **124**, in the implementation shown, the controller **124** includes a user interface **208**, a communication interface **210**, one or more processors **212**, and a memory **214** storing instructions executable by the one or more processors **212** to perform various functions including the disclosed implementations. The user interface **208**, the communication interface **133**, and the memory **214** are electrically and/or communicatively coupled to the one or more processors **212**.

(60) In an implementation, the user interface **208** is adapted to receive input from a user and to

provide information to the user associated with the operation of the system **100** and/or an analysis taking place. The user interface **208** may include a touch screen, a display, a key board, a speaker(s), a mouse, a track ball, and/or a voice recognition system. The touch screen and/or the display may display a graphical user interface (GUI).

(61) In an implementation, the communication interface **210** is adapted to enable communication between the system **100** and a remote system(s) (e.g., computers) via a network(s). The network(s) may include the Internet, an intranet, a local-area network (LAN), a wide-area network (WAN), a coaxial-cable network, a wireless network, a wired network, a satellite network, a digital subscriber line (DSL) network, a cellular network, a Bluetooth connection, a near field communication (NFC) connection, etc. Some of the communications provided to the remote system may be associated with analysis results, imaging data, etc. generated or otherwise obtained by the system **100**. Some of the communications provided to the system **100** may be associated with a fluidics analysis operation, patient records, and/or a protocol(s) to be executed by the system **100**.

(62) The one or more processors **212** and/or the system **100** may include one or more of a processor-based system(s) or a microprocessor-based system(s). In some implementations, the one or more processors **212** and/or the system **100** includes one or more of a programmable processor, a programmable controller, a microprocessor, a microcontroller, a graphics processing unit (GPU), a digital signal processor (DSP), a reduced-instruction set computer (RISC), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a field programmable logic device (FPLD), a logic circuit, and/or another logic-based device executing various functions including the ones described herein.

(63) The memory **214** can include one or more of a semiconductor memory, a magnetically readable memory, an optical memory, a hard disk drive (HDD), an optical storage drive, a solid-state storage device, a solid-state drive (SSD), a flash memory, a read-only memory (ROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), a random-access memory (RAM), a non-volatile RAM (NVRAM) memory, a compact disc (CD), a compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a Blu-ray disk, a redundant array of independent disks (RAID) system, a cache and/or any other storage device or storage disk in which information is stored for any duration (e.g., permanently, temporarily, for extended periods of time, for buffering, for caching).

(64) FIG. 2 is an isometric view a flow cell interface **300** that can be used to implement the flow cell interface **102** of FIG. 1 and an isometric view of the imaging system **114** of FIG. 1. The flow cell interface **300** is similar to the flow cell interface **102** of FIG. 1. The flow cell interface **300** of FIG. 2, however, includes a pair of flow cell supports **104** and a pair of temperature control devices **110** as a result. While two flow cell supports **104** are shown, any number of flow cell supports **104** can be included.

(65) The flow cell interface **300** has a bordering structure **302** adjacent to the flow cell supports **104** and the printed circuit board **136** is shown coupled to the bordering structure **230**. The printed circuit board **136** is positioned between the temperature sensor **132** and the flow cell interface **102** and the metallic layer **138** extends over and covers at least a portion of the surface **140** of the temperature sensor **132**. The metallic layer **138** is a metal cap disposed on the temperature sensor **132** in the implementation shown. The metallic layer **138** can be about a 4×4×4 mm piece of aluminum. The metallic layer **138** may, however, be any other size and be made of another material.

(66) The temperature sensor **132**, the printed circuit board **136**, and/or the metallic layer **138**, can be at ambient temperature because they are spaced from the temperature control device **110** and, thus, spaced from the flow cell support **104**. Less temperature gradients may be present between the temperature sensor **132**, the printed circuit board **136**, and the metallic layer **138** as a result. The infrared sensor **112** may be calibrated more accurately because the components **132**, **136**, **138** are at ambient temperature. Put another way, a position of the components **132**, **136**, **138** on the bordering

structure **302** reduces an amount that the temperature control device **110** affects a temperature of the components **132, 136, 138**.

(67) The metallic layer **138** and the temperature sensor **132** can also have a low total heat content and good thermal conductivity that allows the metallic layer **138** and the temperature sensor **132** to quickly track changes in ambient air temperature to maintain consistent measurements and calibrations. Temperature readings from the temperature sensor **132** can be relied on to be accurate within about 0.15 degree C. for the life of the temperature sensor **132** due to the lack of temperature cycling impacting the thermal bond between the temperature sensor **132** and the metallic layer **138** and the temperature sensor **132** using a semiconductor bandgap.

(68) FIG. **3** is a cross-sectional schematic view of a portion of the flow cell interface **300** of FIG. **2** showing the printed circuit board **136** positioned between the temperature sensor **132** and the flow cell interface **102**. The metallic layer **138** extends over and covers at least a portion of the surface **140** of the temperature sensor **132**.

(69) FIG. **4** is an isometric view of another flow cell interface **400** that can be used to implement the flow cell interface **102** of FIG. **1** and an isometric schematic view of the imaging system **114** of FIG. **1**. The flow cell interface **400** is similar to the flow cell interface **300** of FIG. **2**. The flow cell interface **400** of FIG. **4**, however, includes the temperature sensor **132** coupled to the flow cell support **104** instead of being coupled to the flow cell interface **102**. The temperature sensor **132** is, thus, not spaced from the flow cell support **104** in FIG. **4**. The position of the temperature sensor **132** allows the temperature sensor **132** to measure a temperature of the flow cell support **104** instead of measuring a temperature at the metallic layer **138**.

(70) The infrared sensor **112** (FIG. **1**) measures a first temperature value of the flow cell support **104** during a calibration procedure of the infrared sensor **112** in the implementation shown and the temperature sensor **132** can measure a second temperature value at the flow cell support **104**. The controller **124** of the system **100** of FIG. **1** can calibrate the infrared sensor **112** based on a difference between the first temperature value and the second temperature value in some implementations.

(71) The temperature sensor **132** may be a part of a calibration assembly **402** that includes the temperature sensor **132** and the printed circuit board **136**. The printed circuit board **136** is shown coupled to the temperature sensor **132**. The temperature sensor **132** is positioned between the printed circuit board **136** and the flow cell deck **134**.

(72) FIG. **5** is a cross-sectional schematic view of a portion of the flow cell interface **400** of FIG. **4** showing the temperature sensor **132** coupled to a lower surface **404** of the flow cell support **104**. The printed circuit board **136** is positioned between the temperature sensor **132** and the flow cell interface **102**.

(73) FIG. **6** is a top plan view of a temperature control device **500** that can be used to implement the temperature control device **110** of FIG. **1**. The temperature control device **500** of FIG. **5** includes the first zone **502**, the second zone **504**, and the third zone **506**. Each of the zones **502, 504, 506** has a corresponding thermoelectric cooler **126** and a corresponding resistance temperature detector **128**. The thermoelectric coolers **126** heat and/or cool the corresponding zone **502, 504, 506** and the resistance temperature detectors **128** are positioned to measure a temperature in the corresponding zone **502, 504, 506**. The resistance temperature detectors **128** allow the controller **124** to access the measured temperature values through a feedback loop that allows the commanded temperatures to be achieved. One of the thermoelectric coolers **126** may control a temperature of more than one of the zones **502, 504, 506**, however.

(74) FIGS. **7, 8, 9, and 10** illustrate flowcharts for processes **700, 800, 900, 100** of calibrating the temperature control devices **110, 500** of FIGS. **1, 2, 4, and 6** and the infrared sensors **112** of FIGS. **1, 2, and 4** or any of the other implementations disclosed herein. The order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, combined and/or subdivided into multiple blocks. The processes **700, 800, 900, and/or 1000** can be

performed individually or performed serially in any order.

(75) The process **700** of FIG. **7** begins with the temperature control device **110** of the system **100** being commanded to cause the flow cell support **104** to achieve a temperature value (Block **702**). The temperature control device **110** can be commanded to cause the flow cell support **104** to achieve the temperature value in some implementations by commanding the thermoelectric cooler **126** to cause the flow cell support **104** to achieve the temperature value and the resistance temperature detector **128** measuring the temperature value.

(76) An actual temperature value of the flow cell support **104** is measured using the infrared sensor **112** (Block **704**). The actual temperature value for the flow cell support **104** is measured using the infrared sensor **112** in some implementations while moving the flow cell interface **102** relative to the infrared sensor **112**.

(77) The temperature control device **110** is calibrated based on a difference between the commanded temperature value and the actual temperature value (Block **706**). The temperature control device **110** is calibrated in some implementations in responsive to receiving a command at the controller **124** to calibrate the temperature control device **110**. The controller **124** can receive the command from a remote computing device and/or from user input received at the user interface **208** of the system **100**.

(78) The process **800** of FIG. **8** begins with temperature control device **110** being commanded to cause the flow cell support **104** to achieve a temperature value (Block **802**). The flow cell **108** may be disposed on the flow cell support **104** and no sample and/or reagent may be contained within the flow cell **108**. An actual temperature value of the flow cell **108** disposed on the flow cell support **104** is measured using the infrared sensor **112** (Block **804**) and the commanded temperature value and the actual temperature value are compared (Block **806**). The temperature control device **110** is calibrated based the comparison (Block **808**). The controller **124** can account for a difference between the temperature determined by the resistance temperature detectors **128** at the temperature control device **110** and the temperature determined by the thermoelectric device **110** at the flow cell **108** when calibrating the temperature control device **110**.

(79) The process **900** of FIG. **9** begins with the system **100** being caused to perform at least a portion of an analysis on one or more samples of interest within the flow cell **108** disposed on the flow cell support **104** (Block **902**). The temperature control device **110** is commanded to cause the flow cell support **104** to achieve a temperature value (Block **904**). An actual temperature value of the flow cell **108** disposed on the flow cell support **104** is measured using the infrared sensor **112** (Block **906**) and the commanded temperature value and the actual temperature value are compared (Block **908**). The temperature control device **110** is calibrated based the comparison (Block **910**).

(80) The process **1000** of FIG. **10** begins with the infrared sensor **112** being calibrated (Block **1002**). The infrared sensor **112** can be calibrated by generating data associated with the difference between the first temperature value and the second temperature value and calibrating the infrared sensor **112** with the controller **124** based on the data.

(81) The infrared sensor **112** is calibrated in some implementations by measuring a first temperature value adjacent to the temperature sensor **132** carried by the flow cell interface **102** using the infrared sensor **112**, measuring a second temperature value using the temperature sensor **132**, and calibrating the infrared sensor **112** with the controller **124** based on a difference between the first temperature value and the second temperature value. The infrared sensor **112** is calibrated in other implementation by measuring a first temperature value of the flow cell support **104** using the infrared sensor **112**, measuring a second temperature value at the flow cell support **104** using the temperature sensor **132**, and calibrating the infrared sensor **112** with the controller **124** based on a difference between the first temperature value and the second temperature value. The temperature sensor **132** may be carried by the flow cell support **104** in the second implementation mentioned. The infrared sensor **112** can be calibrated in response to receiving a command at the controller **124** to calibrate the infrared sensor **112**. The command can be received from a remote computing device

and/or from user input received at the user interface **208** of the system **100**.

(82) The temperature control device **110** of the system **100** is commanded to cause the flow cell support **104** to achieve a temperature value (Block **1004**). The temperature control device **110** can be commanded to cause the flow cell support **104** to achieve the temperature value in some implementations by commanding the thermoelectric cooler **126** to cause the flow cell support **104** to achieve the temperature value and the resistance temperature detector **128** measuring the temperature value.

(83) An actual temperature value of the flow cell support **104** is measured using the infrared sensor **112** (Block **1006**). The actual temperature value for the flow cell support **104** is measured using the infrared sensor **112** in some implementations while moving the flow cell interface **102** relative to the infrared sensor **112**.

(84) The temperature control device **110** is calibrated based on a difference between the commanded temperature value and the actual temperature value (Block **1008**). The temperature control device **110** is calibrated in some implementations in responsive to receiving a command at the controller **124** to calibrate the temperature control device **110**. The controller **124** can receive the command from a remote computing device and/or from user input received at the user interface **208** of the system **100**.

(85) The system **100** is caused to perform at least a portion of an analysis on one or more samples of interest within the flow cell **108** disposed on the flow cell support **104** (Block **1010**). The temperature control device **110** is commanded to cause the flow cell support **104** to achieve a second temperature value (Block **1012**). A second actual temperature value of the flow cell **108** disposed on the flow cell support **104** is measured using the infrared sensor **112** (Block **1014**) and the second commanded temperature value and the second actual temperature value are compared (Block **1016**). The temperature control device **110** is recalibrated based the comparison (Block **1018**).

(86) In particular implementations, methods, and optical systems described herein may be used for sequencing nucleic acids. Sequencing-by-synthesis (SBS) protocols are particularly applicable, for example. In SBS, a plurality of fluorescently labeled modified nucleotides are used to sequence dense clusters of amplified DNA (possibly millions of clusters) present on the surface of an optical substrate (e.g., a surface that at least partially defines a channel in a flow cell). The flow cells may contain nucleic acid samples for sequencing where the flow cells are placed within the appropriate flow cell holders. The samples for sequencing can take the form of single nucleic acid molecules that are separated from each other so as to be individually resolvable, amplified populations of nucleic acid molecules in the form of clusters or other features, or beads that are attached to one or more molecules of nucleic acid. The nucleic acids can be prepared such that they comprise an oligonucleotide primer adjacent to an unknown target sequence. To initiate the first SBS sequencing cycle, one or more differently labeled nucleotides, and DNA polymerase, etc., can be flowed into/through the flow cell by a fluid flow subsystem (not shown). Either a single type of nucleotide can be added at a time, or the nucleotides used in the sequencing procedure can be specially designed to possess a reversible termination property, thus allowing each cycle of the sequencing reaction to occur simultaneously in the presence of several types of labeled nucleotides (e.g. A, C, T, G). The nucleotides can include detectable label moieties such as fluorophores. Where the four nucleotides are mixed together, the polymerase is able to select the correct base to incorporate and each sequence is extended by a single base. Nonincorporated nucleotides can be washed away by flowing a wash solution through the flow cell. One or more lasers may excite the nucleic acids and induce fluorescence. The fluorescence emitted from the nucleic acids is based upon the fluorophores of the incorporated base, and different fluorophores may emit different wavelengths of emission light. A deblocking reagent can be added to the flow cell to remove reversible terminator groups from the DNA strands that were extended and detected. The deblocking reagent can then be washed away by flowing a wash solution through the flow cell. The

flow cell is then ready for a further cycle of sequencing starting with introduction of a labeled nucleotide as set forth above. The fluidic and detection steps can be repeated several times to complete a sequencing run. Example sequencing methods are described, for example, in Bentley et al., Nature 456:53-59 (2008), WO 04/018497; U.S. Pat. No. 7,057,026; WO 91/06678; WO 07/123,744; U.S. Pat. Nos. 7,329,492; 7,211,414; 7,315,019; 7,405,281, and US 2008/0108082, each of which is incorporated herein by reference.

(87) Optical systems described herein may also be used to scan samples that include microarrays. A microarray may include a population of different probe molecules that are attached to one or more substrates such that the different probe molecules can be differentiated from each other according to relative location. An array can include different probe molecules, or populations of the probe molecules, that are each located at a different addressable location on a substrate. Alternatively, a microarray can include separate optical substrates, such as beads, each bearing a different probe molecule, or population of the probe molecules, that can be identified according to the locations of the optical substrates on a surface to which the substrates are attached or according to the locations of the substrates in a liquid. Exemplary arrays in which separate substrates are located on a surface include, without limitation, a BeadChip Array available from Illumina, Inc. (San Diego, Calif.) or others including beads in wells such as those described in U.S. Pat. Nos. 6,266,459, 6,355,431, 6,770,441, 6,859,570, and 7,622,294; and PCT Publication No. WO 00/63437, each of which is hereby incorporated by reference. Other arrays having particles on a surface include those set forth in US 2005/0227252; WO 05/033681; and WO 04/024328, each of which is hereby incorporated by reference.

(88) An example apparatus, comprising: a flow cell interface including a flow cell support; a temperature control device for the flow cell support; an infrared sensor; and a controller to command the temperature control device to cause the flow cell support to achieve a temperature value, cause the infrared sensor to measure an actual temperature value of the flow cell support, and calibrate the temperature control device based on a difference between the commanded temperature value and the actual temperature value.

(89) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein the temperature control device comprises a thermoelectric cooler.

(90) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein the temperature control device further comprises a resistance temperature detector.

(91) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein the controller is to calibrate the temperature control device by calibrating the resistance temperature detector.

(92) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising an imaging system, the infrared sensor coupled to the imaging system.

(93) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising a bracket coupling the infrared sensor to the imaging system.

(94) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein the controller is further to calibrate the infrared sensor.

(95) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising a temperature sensor; wherein the infrared sensor is to measure a first temperature value adjacent to the temperature sensor; the temperature sensor is to measure a second temperature value; and the controller is to calibrate the infrared sensor based on a difference between the first temperature value and the second temperature value.

(96) The apparatus of any one or more of the preceding embodiments and/or any one or more of the

embodiments disclosed below, wherein the flow cell interface further comprises a flow cell deck carrying the flow cell support, the temperature sensor coupled to the flow cell deck.

(97) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising a printed circuit board coupled to the temperature sensor, the printed circuit board positioned between the temperature sensor and the flow cell deck.

(98) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising a metallic layer covering a surface of the temperature sensor.

(99) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein the metallic layer and a surface of the flow cell support have substantially similar emissivity.

(100) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein the temperature sensor and the metallic layer are spaced from the flow cell support.

(101) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein the metallic layer comprises aluminum.

(102) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising thermally conductive epoxy coupling the temperature sensor and the metallic layer.

(103) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising a temperature sensor; wherein the infrared sensor is to measure a first temperature value of the flow cell support; the temperature sensor is to measure a second temperature value at the flow cell support; and the controller is to calibrate the infrared sensor based on a difference between the first temperature value and the second temperature value.

(104) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein the temperature sensor is coupled to the flow cell support.

(105) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising a printed circuit board coupled to the temperature sensor, the temperature sensor positioned between the flow cell interface and the printed circuit board.

(106) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein a surface of the flow cell support comprises a metallic layer.

(107) The apparatus of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein the temperature sensor comprises a digital temperature sensor.

(108) A method, comprising: commanding a temperature control device of a system to cause a flow cell support to achieve a temperature value; measuring an actual temperature value of the flow cell support using an infrared sensor; and calibrating the temperature control device based on a difference between the commanded temperature value and the actual temperature value.

(109) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein commanding the temperature control device to cause the flow cell support to achieve the temperature value comprises: commanding a thermoelectric cooler to cause the flow cell support to achieve the temperature value; and measuring the temperature value with one or more resistance temperature detectors.

(110) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein measuring the actual temperature value for the flow cell

support using the infrared sensor comprises measuring the actual temperature value for the flow cell support using the infrared sensor while moving the flow cell interface relative to the infrared sensor.

(111) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising: commanding the temperature control device to cause the flow cell support to achieve a second temperature value; measuring a second actual temperature value of a flow cell disposed on the flow cell support using the infrared sensor; comparing the second commanded temperature value and the second actual temperature value; and recalibrating the temperature control device based the comparison.

(112) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising: causing the system to perform at least a portion of an analysis on one or more samples of interest within a flow cell disposed on the flow cell support; and commanding the temperature control device to cause the flow cell support to achieve a second temperature value; measuring a second actual temperature value of a flow cell disposed on the flow cell support using the infrared sensor; comparing the second commanded temperature value and the second actual temperature value; and recalibrating the temperature control device based the comparison.

(113) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein calibrating the temperature control device is responsive to receiving a command at the controller to calibrate the temperature control device.

(114) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein receiving the command at the controller to calibrate the temperature control device comprises receiving the command from a remote computing device.

(115) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein receiving the command at the controller to calibrate the temperature control device comprises receiving user input at a user interface of the system.

(116) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising calibrating the infrared sensor.

(117) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein calibrating the infrared sensor comprises: measuring a first temperature value adjacent to a temperature sensor carried by the flow cell interface using the infrared sensor; measuring a second temperature value using the temperature sensor; and calibrating the infrared sensor with a controller based on a difference between the first temperature value and the second temperature value.

(118) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein calibrating the infrared sensor comprises: measuring a first temperature value of the flow cell support using the infrared sensor; measuring a second temperature value at the flow cell support using a temperature sensor, the temperature sensor being carried by the flow cell support; and calibrating the infrared sensor with the controller based on a difference between the first temperature value and the second temperature value.

(119) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein calibrating the infrared sensor is responsive to receiving a command at the controller to calibrate the infrared sensor.

(120) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein receiving the command at the controller to calibrate the infrared sensor comprises receiving the command from a remote computing device.

(121) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein receiving the command at the controller to calibrate the infrared sensor comprises receiving user input at a user interface of the system.

(122) The method of any one or more of the preceding embodiments and/or any one or more of the

embodiments disclosed below, wherein calibrating the infrared sensor comprises generating data associated with the difference between the first temperature value and the second temperature value and calibrating the infrared sensor with the controller based on the data.

(123) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein commanding the temperature control device of the system to cause the flow cell support to achieve the temperature value comprises commanding a first zone of the temperature control device of the system to cause a first zone of the flow cell support to achieve the temperature value and wherein measuring the actual temperature value of the flow cell support using the infrared sensor comprises measuring an actual temperature value of the first zone of the flow cell support.

(124) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, wherein calibrating the temperature control device based on a difference between the commanded temperature value and the actual temperature value comprises calibrating a resistance temperature detector of the first zone of the temperature control device based on a difference between the commanded temperature value and the actual temperature value.

(125) The method of any one or more of the preceding embodiments and/or any one or more of the embodiments disclosed below, further comprising 1) commanding a second zone of the temperature control device of the system to cause a second zone of the flow cell support to achieve a second temperature value; 2) measuring an actual temperature value of the second zone of flow cell support using the infrared sensor; and 3) calibrating a resistance temperature detector of the second zone of the temperature control device based on a difference between the commanded temperature value and the actual temperature value.

(126) A method, comprising: commanding a temperature control device of a system to cause a flow cell support to achieve a temperature value; measuring a plurality of actual temperature value of the flow cell support using an infrared sensor; determining an average actual temperature value based on the plurality of actual temperature values; and calibrating the temperature control device based on a difference between the commanded temperature value and the average actual temperature value.

(127) The foregoing description is provided to enable a person skilled in the art to practice the various configurations described herein. While the subject technology has been particularly described with reference to the various figures and configurations, it should be understood that these are for illustration purposes only and should not be taken as limiting the scope of the subject technology.

(128) As used herein, an element or step recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural of said elements or steps, unless such exclusion is explicitly stated. Furthermore, references to “one implementation” are not intended to be interpreted as excluding the existence of additional implementations that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, implementations “comprising,” “including,” or “having” an element or a plurality of elements having a particular property may include additional elements whether or not they have that property. Moreover, the terms “comprising,” “including,” “having,” or the like are interchangeably used herein.

(129) The terms “substantially,” “approximately,” and “about” used throughout this Specification are used to describe and account for small fluctuations, such as due to variations in processing. In certain implementations, they can refer to less than or equal to $\pm 5\%$, such as less than or equal to $\pm 2\%$, such as less than or equal to $\pm 1\%$, such as less than or equal to $\pm 0.5\%$, such as less than or equal to $\pm 0.2\%$, such as less than or equal to $\pm 0.1\%$, such as less than or equal to $\pm 0.05\%$.

(130) There may be many other ways to implement the subject technology. Various functions and elements described herein may be partitioned differently from those shown without departing from the scope of the subject technology. Various modifications to these implementations may be readily apparent to those skilled in the art, and generic principles defined herein may be applied to other

implementations. Thus, many changes and modifications may be made to the subject technology, by one having ordinary skill in the art, without departing from the scope of the subject technology. For instance, different numbers of a given module or unit may be employed, a different type or types of a given module or unit may be employed, a given module or unit may be added, or a given module or unit may be omitted.

(131) Underlined and/or italicized headings and subheadings are used for convenience only, do not limit the subject technology, and are not referred to in connection with the interpretation of the description of the subject technology. All structural and functional equivalents to the elements of the various implementations described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and intended to be encompassed by the subject technology. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the above description.

(132) It should be appreciated that all combinations of the foregoing concepts and additional concepts discussed in greater detail below (provided such concepts are not mutually inconsistent) are contemplated as being part of the subject matter disclosed herein. In particular, all combinations of claimed subject matter appearing at the end of this disclosure are contemplated as being part of the subject matter disclosed herein.

Claims

1. An apparatus, comprising: a flow cell interface including a flow cell support adapted to have a flow cell positioned thereon; a temperature control device for the flow cell support; an infrared sensor; a stage assembly configured to move the flow cell interface relative to the infrared sensor; and a controller to command the temperature control device to cause the flow cell support to achieve a temperature value, cause the infrared sensor to measure an actual temperature value of the flow cell support, and calibrate the temperature control device based on a difference between the commanded temperature value and the actual temperature value.
2. The apparatus of claim 1, wherein the temperature control device comprises a thermoelectric cooler.
3. The apparatus of claim 2, wherein the temperature control device further comprises a resistance temperature detector.
4. The apparatus of claim 3, wherein the controller is to calibrate the temperature control device by calibrating the resistance temperature detector.
5. The apparatus of claim 1, further comprising an imaging system, the infrared sensor coupled to the imaging system.
6. The apparatus of claim 1, wherein the controller is further to calibrate the infrared sensor.
7. The apparatus of claim 1, further comprising a temperature sensor; wherein the infrared sensor is to measure a first temperature value adjacent to the temperature sensor; the temperature sensor is to measure a second temperature value; and the controller is to calibrate the infrared sensor based on a difference between the first temperature value and the second temperature value.
8. The apparatus of claim 7, wherein the flow cell interface further comprises a flow cell deck carrying the flow cell support, the temperature sensor coupled to the flow cell deck.
9. The apparatus of claim 8, further comprising a printed circuit board coupled to the temperature sensor, the printed circuit board positioned between the temperature sensor and the flow cell deck.
10. The apparatus of claim 7, further comprising a metallic layer covering a surface of the temperature sensor.
11. The apparatus of claim 1, further comprising a temperature sensor; wherein the infrared sensor is to measure a first temperature value of the flow cell support; the temperature sensor is to measure a second temperature value at the flow cell support; and the controller is to calibrate the

infrared sensor based on a difference between the first temperature value and the second temperature value.

12. The apparatus of claim 11, further comprising a printed circuit board coupled to the temperature sensor, the temperature sensor positioned between the flow cell interface and the printed circuit board.

13. The apparatus of claim 1, wherein a surface of the flow cell support comprises a metallic layer.

14. An apparatus, comprising: a flow cell interface including a flow cell support adapted to have a flow cell positioned thereon; a temperature control device for the flow cell support; an infrared sensor; a stage assembly configured to move the flow cell interface relative to the infrared sensor; and a controller to command the temperature control device to cause the flow cell support to achieve a temperature value, and cause the infrared sensor to measure a plurality of actual temperature values of the flow cell support.

15. The apparatus of claim 14, wherein the controller is to generate a heat map based on the plurality of actual temperature values measured.

16. The apparatus of claim 15, wherein the controller is to diagnosis an error based on the heat map.

17. The apparatus of claim 15, wherein the controller is to calibrate the temperature control device based on the heat map.

18. The apparatus of claim 15, wherein the heat map comprises a 2-D heat map.

19. An apparatus, comprising: a flow cell interface including a flow cell support adapted to support a flow cell; a temperature control device for the flow cell support; an infrared sensor; a temperature sensor; a stage assembly configured to move the flow cell interface relative to the infrared sensor; and a controller, wherein the infrared sensor is to measure a first temperature value adjacent to the temperature sensor; the temperature sensor is to measure a second temperature value; and the controller is to calibrate the infrared sensor based on a difference between the first temperature value and the second temperature value.

20. The apparatus of claim 19, further comprising a flow cell carrying the temperature sensor.

21. The apparatus of claim 20, wherein the flow cell comprises a mock flow cell.

22. An apparatus, comprising: a flow cell interface including a flow cell support adapted to have a flow cell positioned thereon; a temperature control device for the flow cell support; an infrared sensor; and a controller to command the temperature control device to cause the flow cell support to achieve a temperature value, cause the infrared sensor to measure an actual temperature value of the flow cell support, and calibrate the temperature control device based on a difference between the commanded temperature value and the actual temperature value, wherein the flow cell interface or the infrared sensor are configured to move relative to the other of the flow cell interface or the infrared sensor.

23. The apparatus of claim 22, further comprising a stage assembly configured to move the flow cell interface relative to the infrared sensor.
